Author: Apparently ε need not be specified, atleast at the beginning.

Reader: OK. Therefore, for an arbitrary positive value of ε, I have to find N such that for all $n > N$ the following inequality holds $\boxed{\frac{1}{n+1} < \varepsilon}$.

Author: Quite correct. Go on.

Reader: The inequality can be rewritten in the form $\boxed{n > \frac{1}{\varepsilon} - 1}$

It follows that the unknown N may be identified as an integral part of $\frac{1}{\varepsilon} - 1$. Apparently, for all $n > N$ the inequality in question will hold.

Author: That's right. Let, for example, $\varepsilon = 0.01$.

Reader: Then $N = \frac{1}{\varepsilon} - 1 = \frac{1}{0.01} - 1 = 99$

Author: Let $\varepsilon = 0.001$.

Reader: Then $N = \frac{1}{\varepsilon} - 1 = \frac{1}{0.001} - 1 = 999$

Author: Let $\varepsilon = 0.00015$

Reader: Then $\frac{1}{\varepsilon} - 1 = 6665.(6)$, so that $N = 6665$.

Author: It is quite evident that for any ε (no matter how small) we can find a corresponding N.

As to proving that the limits of sequence (7) and (9) are zero, we shall leave it to the reader as an exercise.

Son: **Claim:** Seq 7 $\longrightarrow \lim_{n \to \infty} \frac{1}{n} = 0$

Proof: Take any arbitrary positive number ε.

Suppose, $\forall K \in R$, $\left|\frac{1}{K} - 0\right| > \varepsilon \Rightarrow \frac{1}{K} > \varepsilon \Rightarrow \boxed{\frac{1}{\varepsilon} > K}$

We know that the set of real numbers cannot have an upper bound.

$\therefore \exists K \in R$ s.t. $\left|\frac{1}{K} - 0\right| < \varepsilon. \Rightarrow \frac{1}{K} < \varepsilon \Rightarrow \frac{1}{\varepsilon} < K. \therefore \forall x \in R, x \geq K,$

$\frac{1}{\varepsilon} < x \Rightarrow \left|\frac{1}{x} - 0\right| < \varepsilon. \therefore \forall \varepsilon > 0, \exists K \in R$ s.t. $\forall x \in R, x \geq K,$

$\left|\frac{1}{x} - 0\right| < \varepsilon$ $\square$